Question 1 (Second-Order Corollary)

Problem Statement

If $u = \frac{x^3y+y^3x}{y-x}$, prove that:

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 6\left(\frac{x^3y+y^3x}{y-x}\right)$$

Step-by-Step Solution

Step 1: Verify Homogeneity and Find the Degree (n)

Define the function explicitly:

$$f(x,y) = \frac{x^3y+y^3x}{y-x}$$

Substitute $x \to xt$ and $y \to yt$ into the function:

$$f(xt,yt) = \frac{(xt)^3(yt)+(yt)^3(xt)}{(yt)-(xt)}$$

Expand the power terms in both components of the numerator:

$$f(xt,yt) = \frac{(x^3t^3 \cdot yt)+(y^3t^3 \cdot xt)}{yt-xt}$$

$$f(xt,yt) = \frac{x^3y \cdot t^4+y^3x \cdot t^4}{yt-xt}$$

Factor out t^4 from the numerator and t^1 from the denominator:

$$f(xt,yt) = \frac{t^4(x^3y+y^3x)}{t^1(y-x)}$$

Simplify the powers of t using the exponent quotient rule $\frac{t^4}{t^1} = t^{4-1} = t^3$:

$$f(xt,yt) = t^3\left(\frac{x^3y+y^3x}{y-x}\right)$$

Substitute the original definition of u back into the equation:

$$f(xt,yt) = t^3u$$

This confirms that u is a homogeneous function with a degree of $n = 3$.

Step 2: Apply the Second-Order Extension of Euler's Theorem

The second-order corollary formula for a homogeneous function of degree n states:

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

Substitute the value $n = 3$ directly into the formula:

$$\text{RHS} = 3(3-1)u$$

$$\text{RHS} = 3(2)u = 6u$$

Substitute the original algebraic definition of u back into the right-hand expression:

$$\text{RHS} = 6\left(\frac{x^3y + y^3x}{y - x}\right)$$

Equating the left-hand side operator to our calculated right-hand side gives:

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 6\left(\frac{x^3y + y^3x}{y - x}\right) = \text{RHS}$$

Hence, proved.